Java 21 Architecture & Virtual Threads

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DPU Course Material

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Virtual Thread Per Task Executor

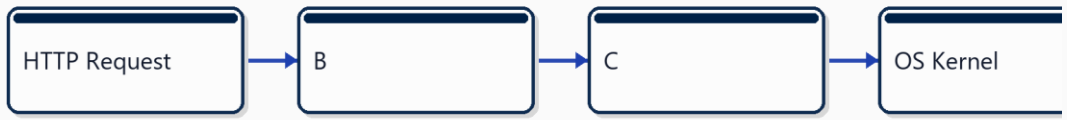
- • Million thread capability without memory exhaustion
- • Automated mounting on underlying carrier pool
- • Transparent non-blocking I/O

```
                    // JAVA
try (var pool = Executors.newVirtualThreadPerTaskExecutor...
    IntStream.range(0, 10).forEach(idx -> {
        pool.submit(() -> {
            Thread.sleep(Duration.ofMillis(50));
            return idx * 2;
        });
    });
}
```


multiplexing

Carrier Thread Scheduling Architecture

- • Virtual threads yield during blocking syscalls
- • ForkJoinPool manages carrier workers efficiently
- • Deterministic stack unmounting in heap memory



Production Virtual Thread Guidelines

- • Never pool virtual threads; spawn per task
- • Avoid pinning with synchronized keyword
- • Migrate to ReentrantLock for critical sections
- • Inspect heap footprint with JFR telemetry
- *Exercise Reference: exercise_1_virtual_threads*